

# TE and TW — two straight upper chords

One long sloping run beneath the solar roof.

One long horizontal run beneath the separate lightweight hip cap.

Horizontal chord axis: 224.25 in above datum (18 ft 8¼ in).

Horizontal run: 130.24 in (approximately 10 ft 10¼ in).

The existing sloping runs are retained; each meets its horizontal run at a single knee.

The hip-cap outline and its roof-load allowance remain unchanged.

Whole-frame second-order analysis: solved.

Maximum partial member-check ratio: 0.835

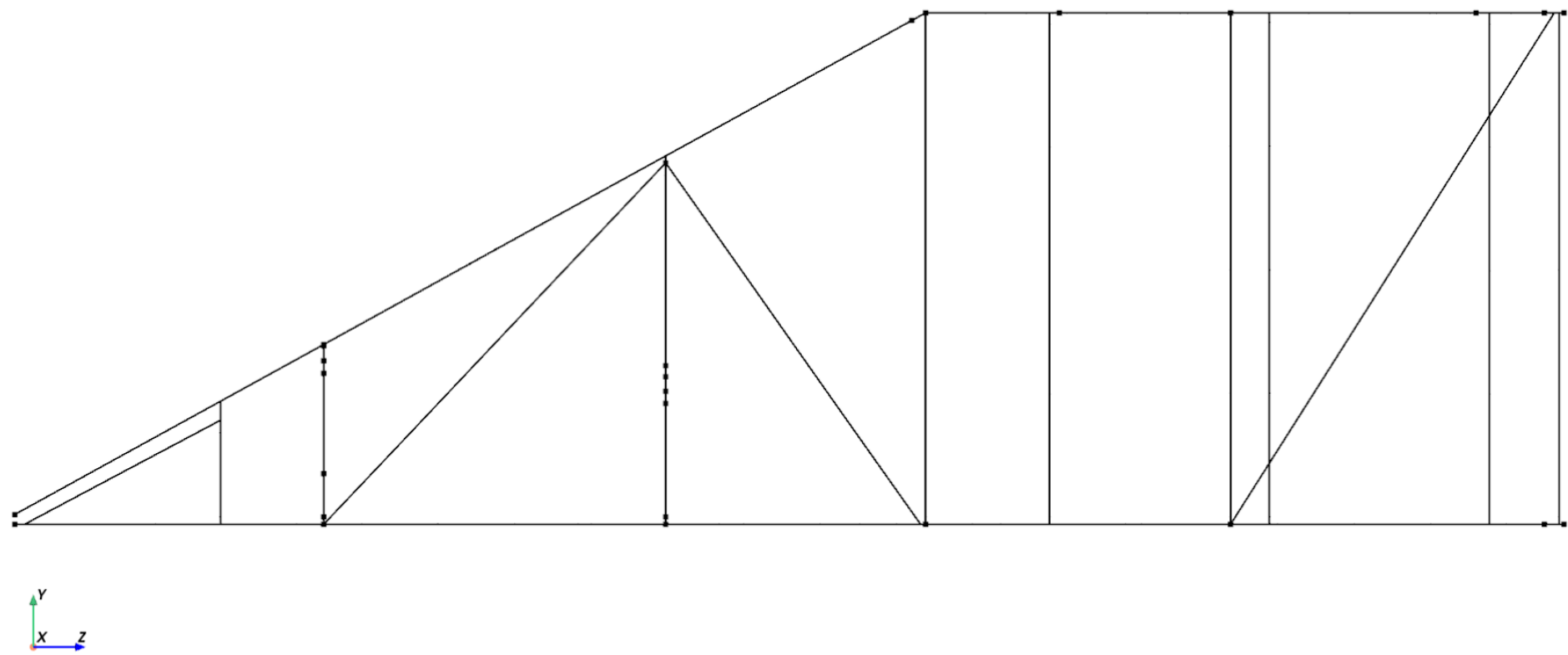
Maximum modeled floor-support movement: 0.54 in

Independent verification: 10 networks, 350 load-case solves.

Primary steel: 9,349 lb; stock sizes unchanged.

The separate sheet-metal hip-cap frame is not sized here. Suggested 1 in, 11-gauge tubing is not an approved section. Existing load allowances include the cap; verify its eventual takeoff. Preliminary model: connection detailing, bracing, foundations and site/code loads remain to verify. All diagrams are rendered directly from the PyNite model, including connection-offset members.

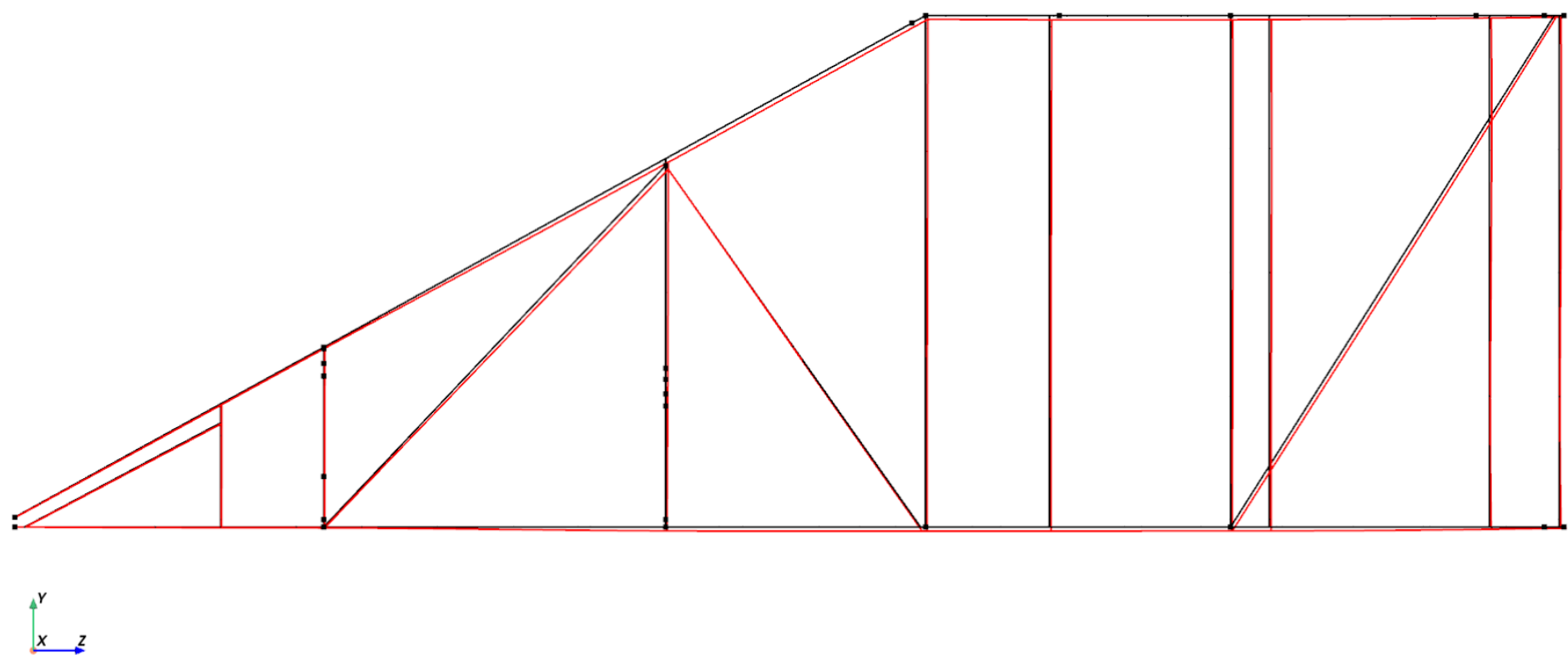
# TE complete PyNite solved network



All physical members AND connection-offset elements shown. Nodes are real solver nodes.  
Boundary glyphs represent imposed global-frame movements, not additional ground supports.

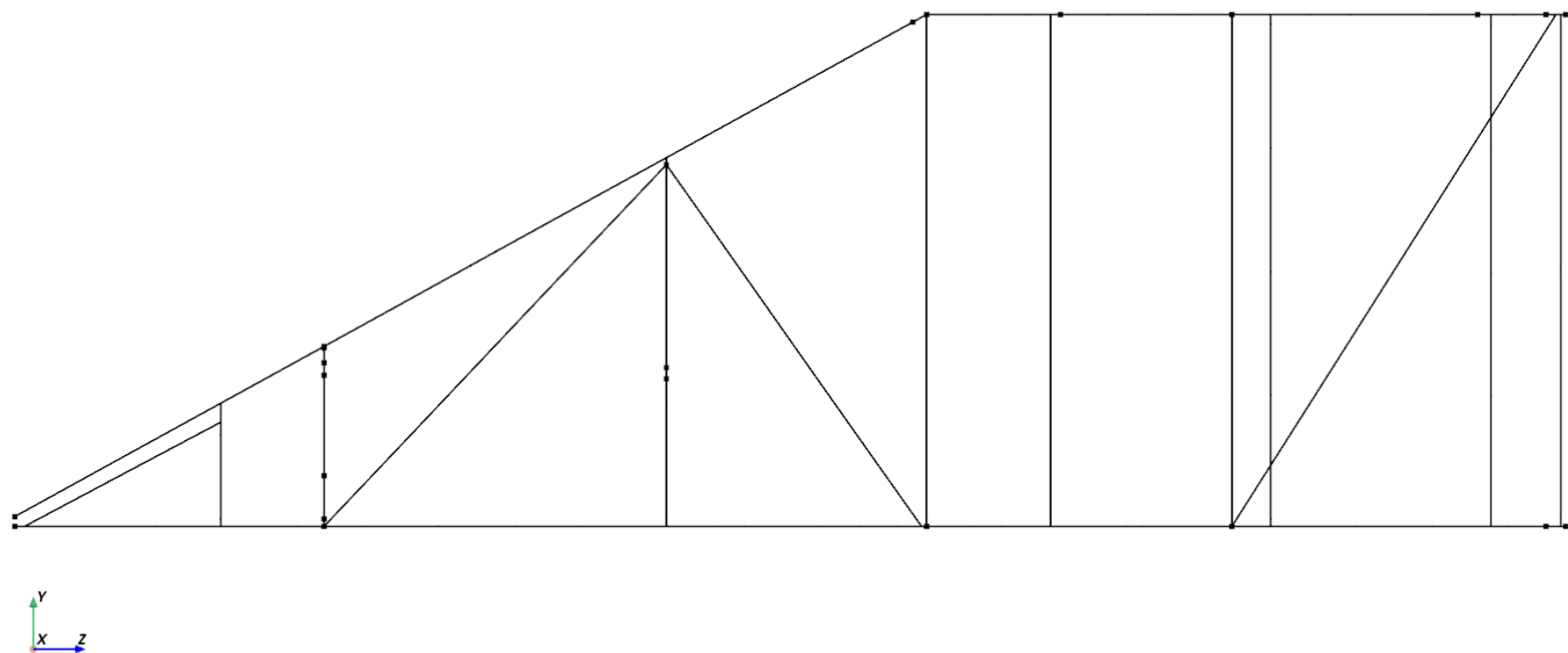
# TE complete PyNite solved network

RED: displacement x20 service\_patches



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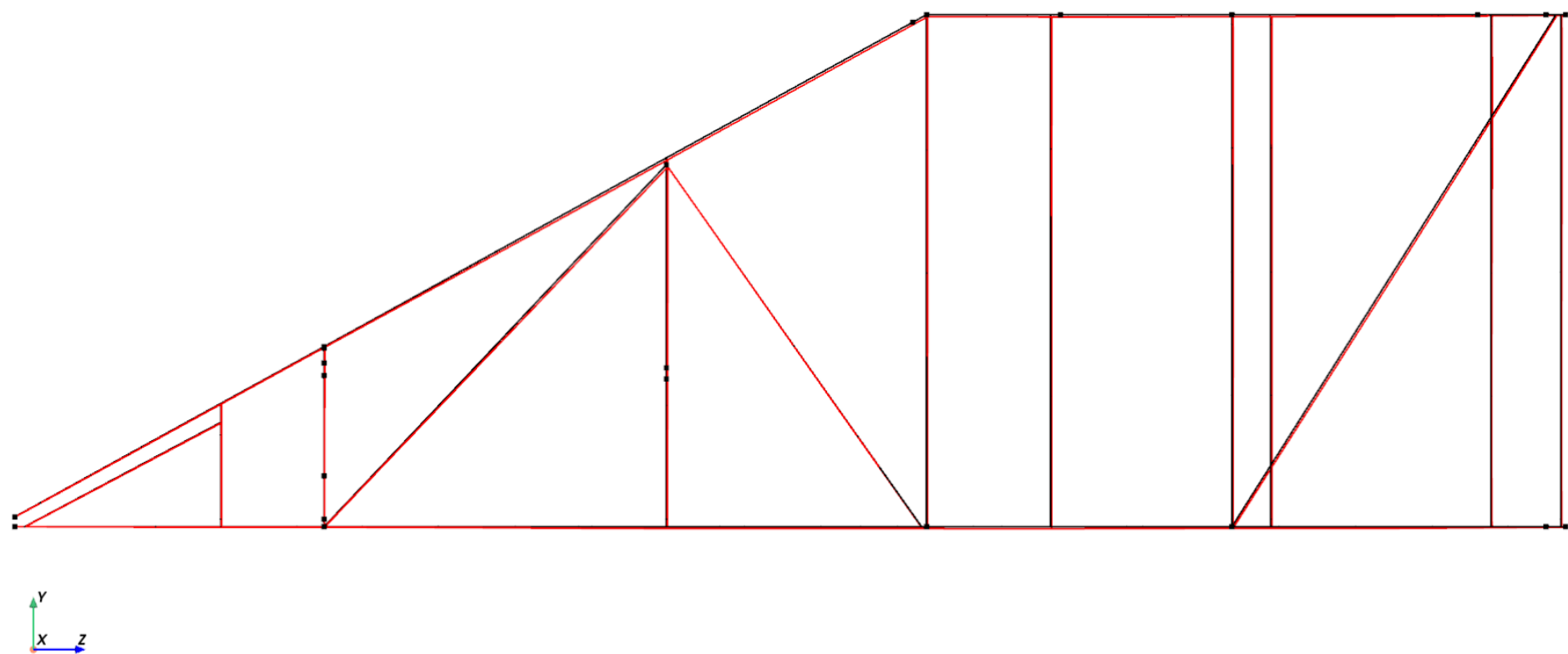
TE roof\_first PyNite solved network



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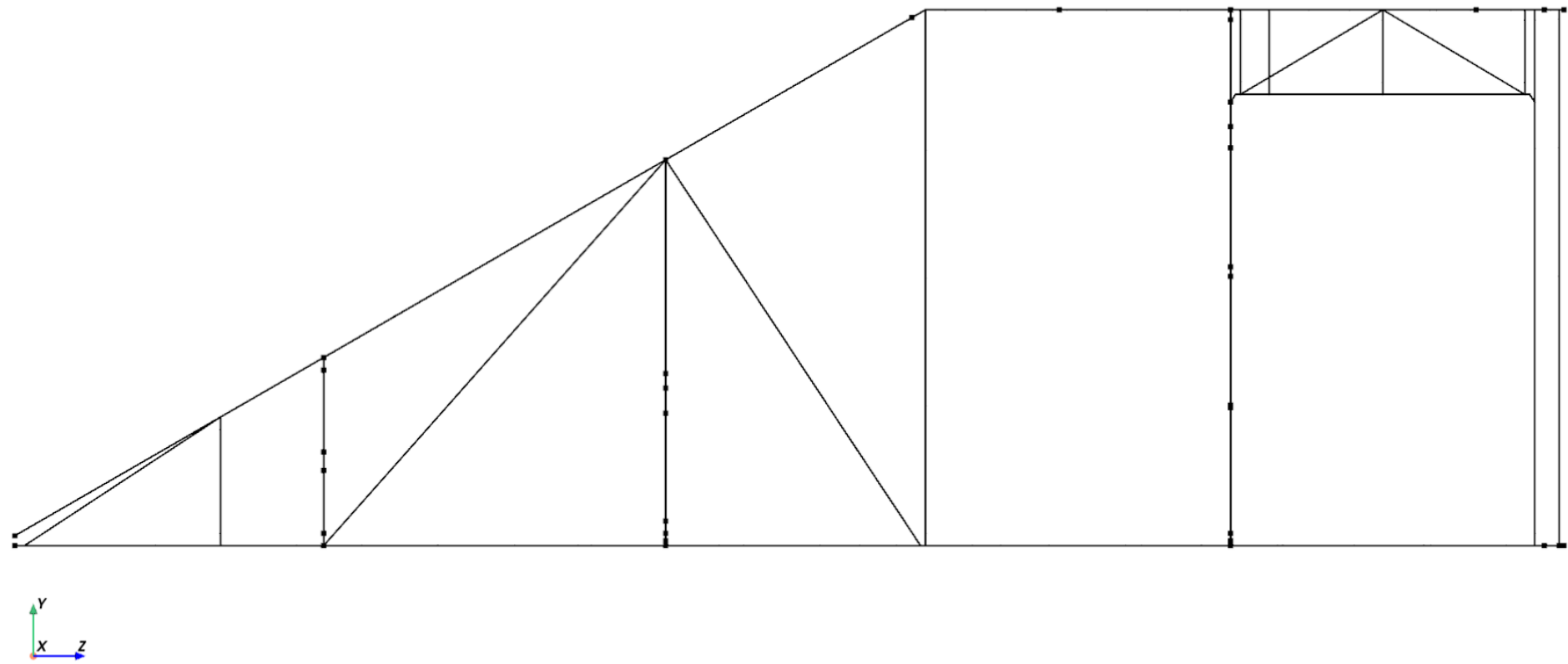
TE roof\_first PyNite solved network

RED: displacement x20 roof\_first\_service



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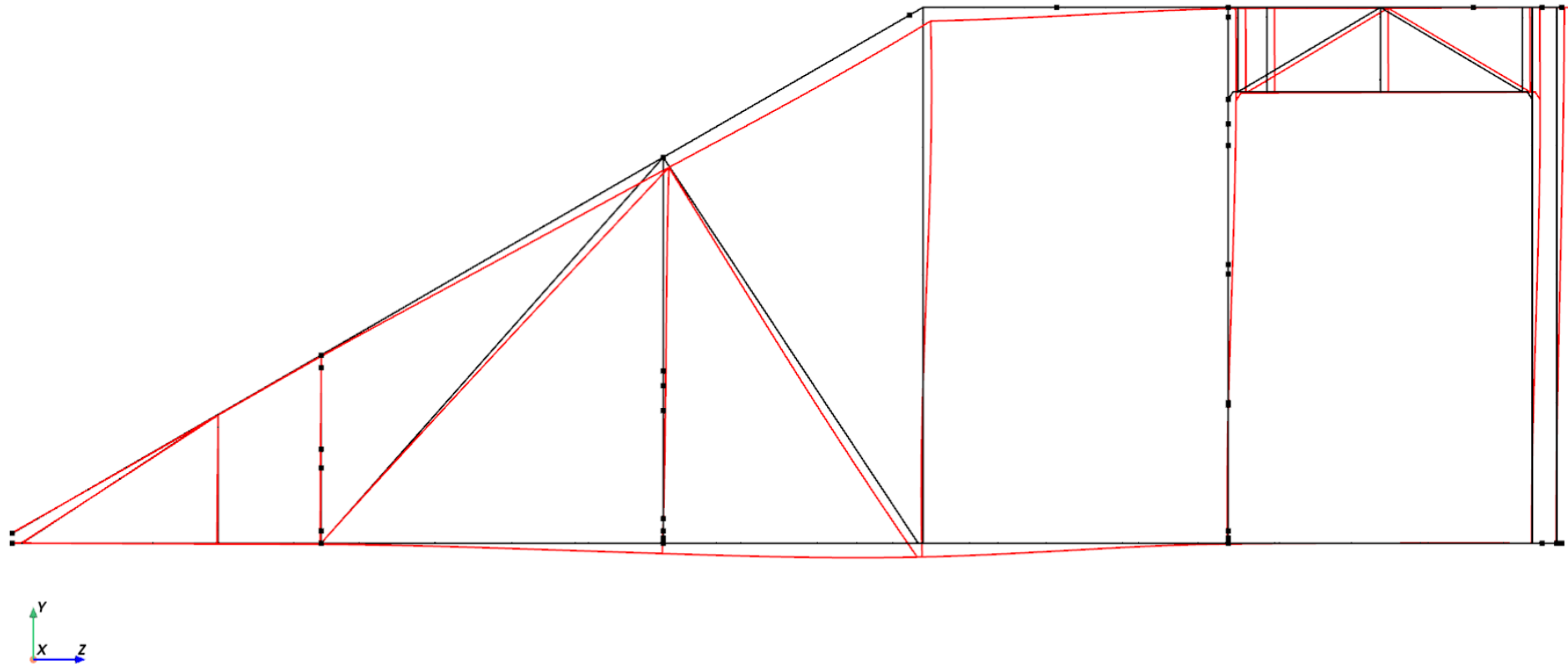
# TW complete PyNite solved network



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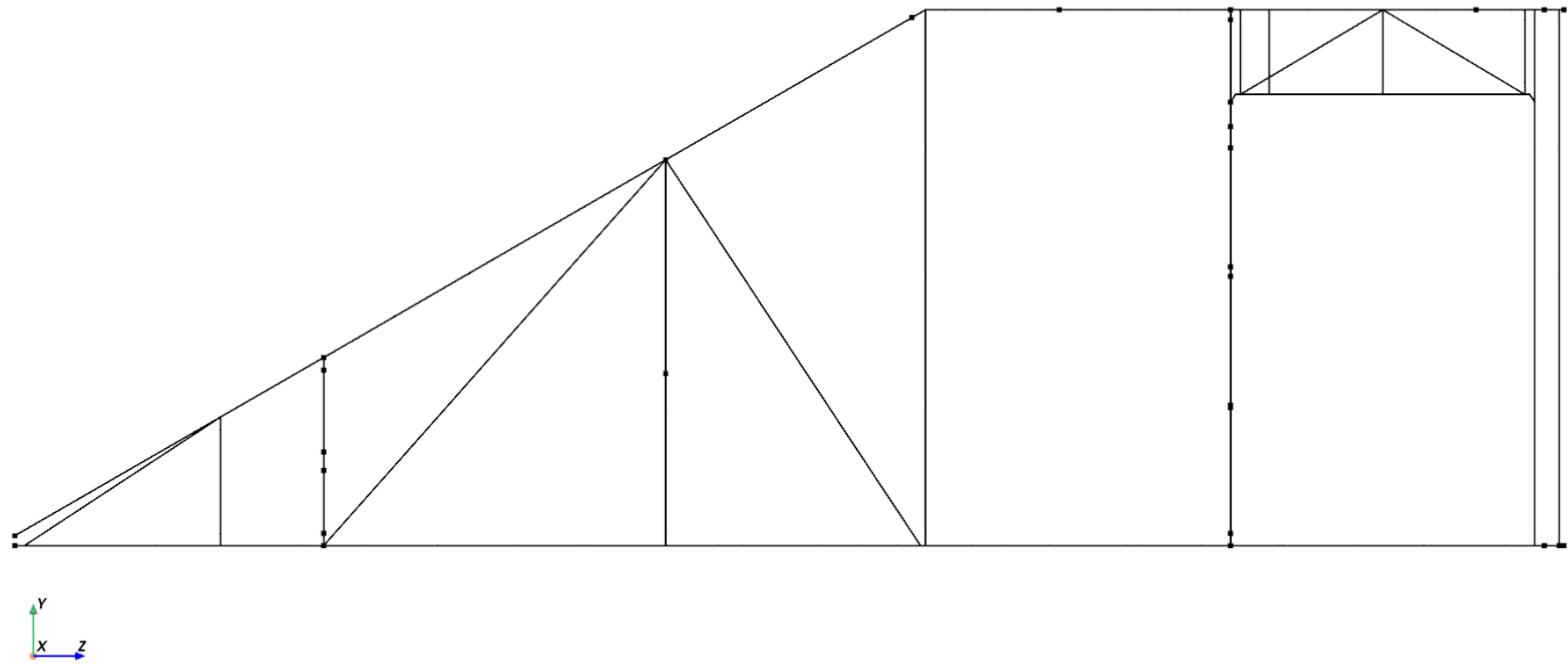
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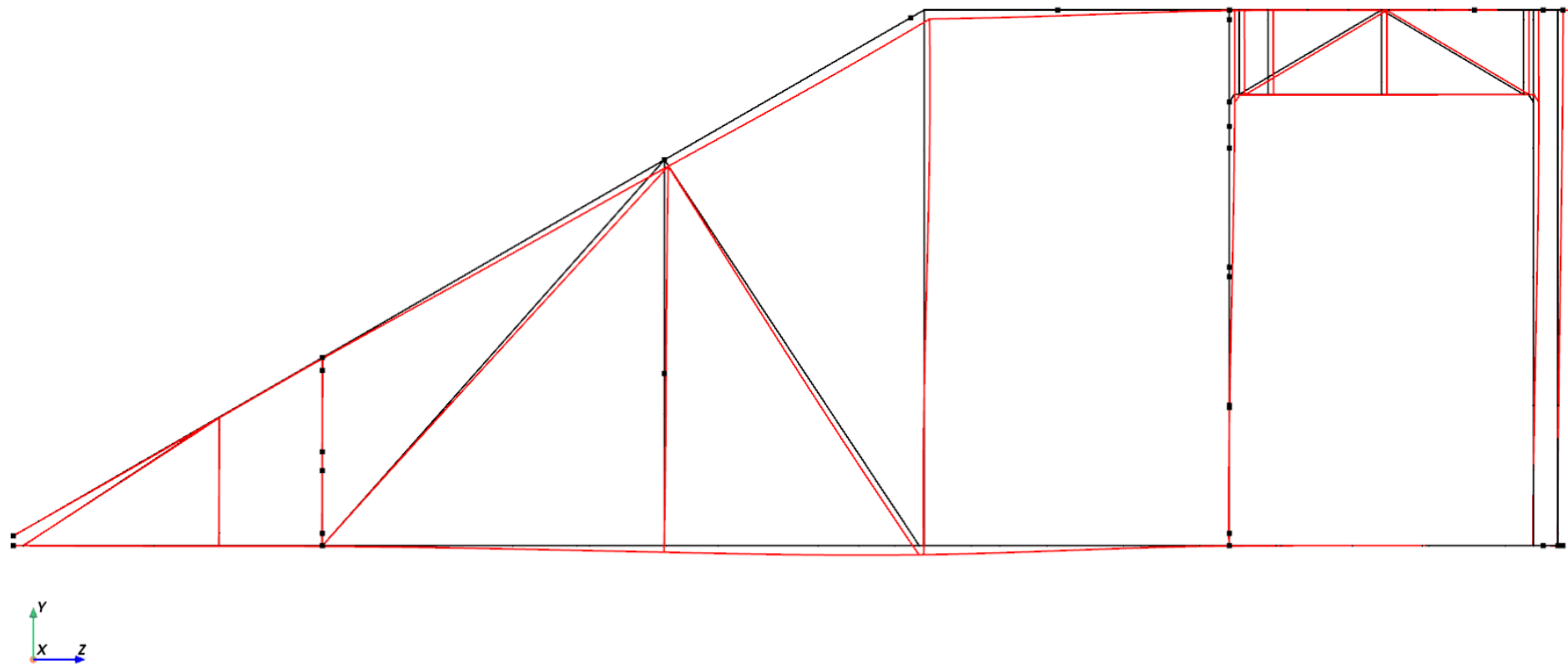
TW roof\_first PyNite solved network



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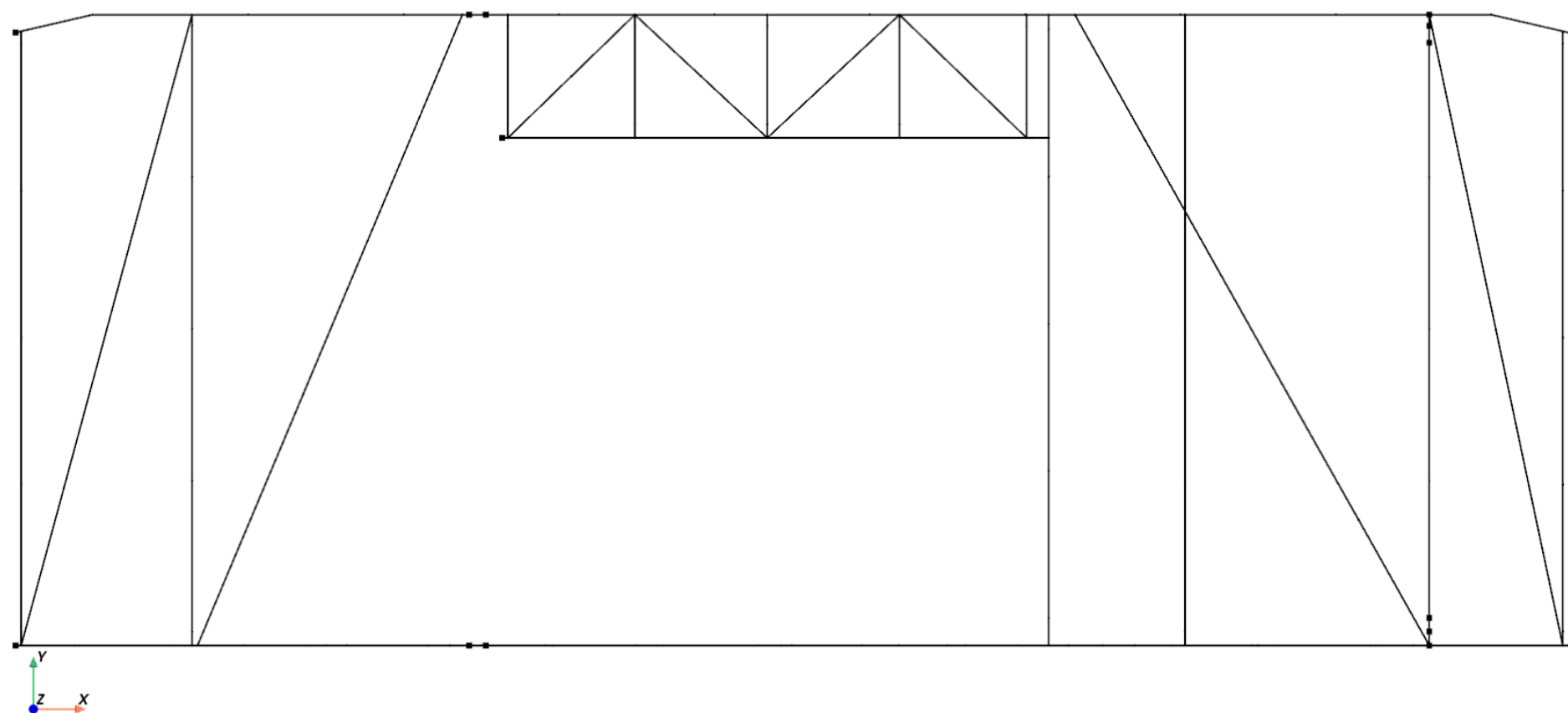
TW roof\_first PyNite solved network

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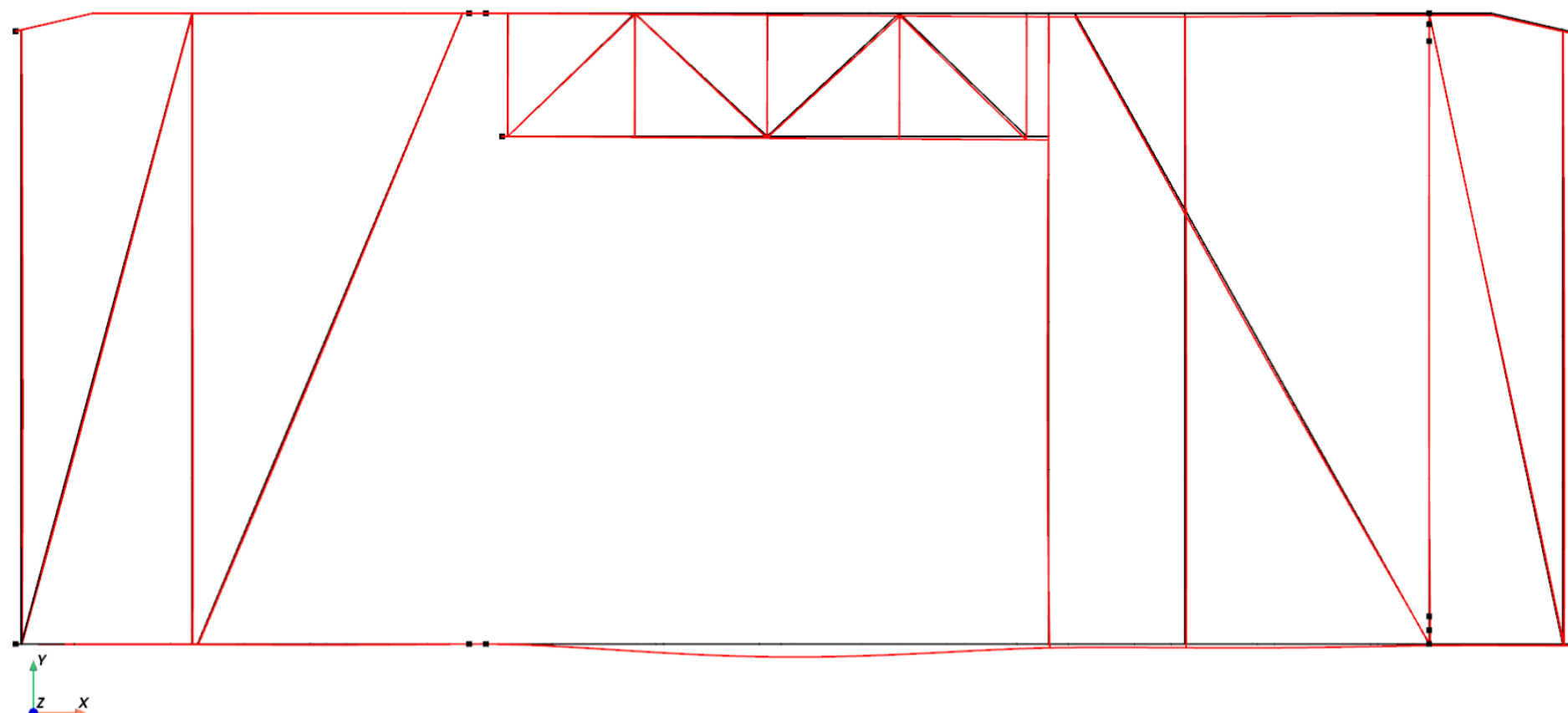
# TN complete PyNite solved network



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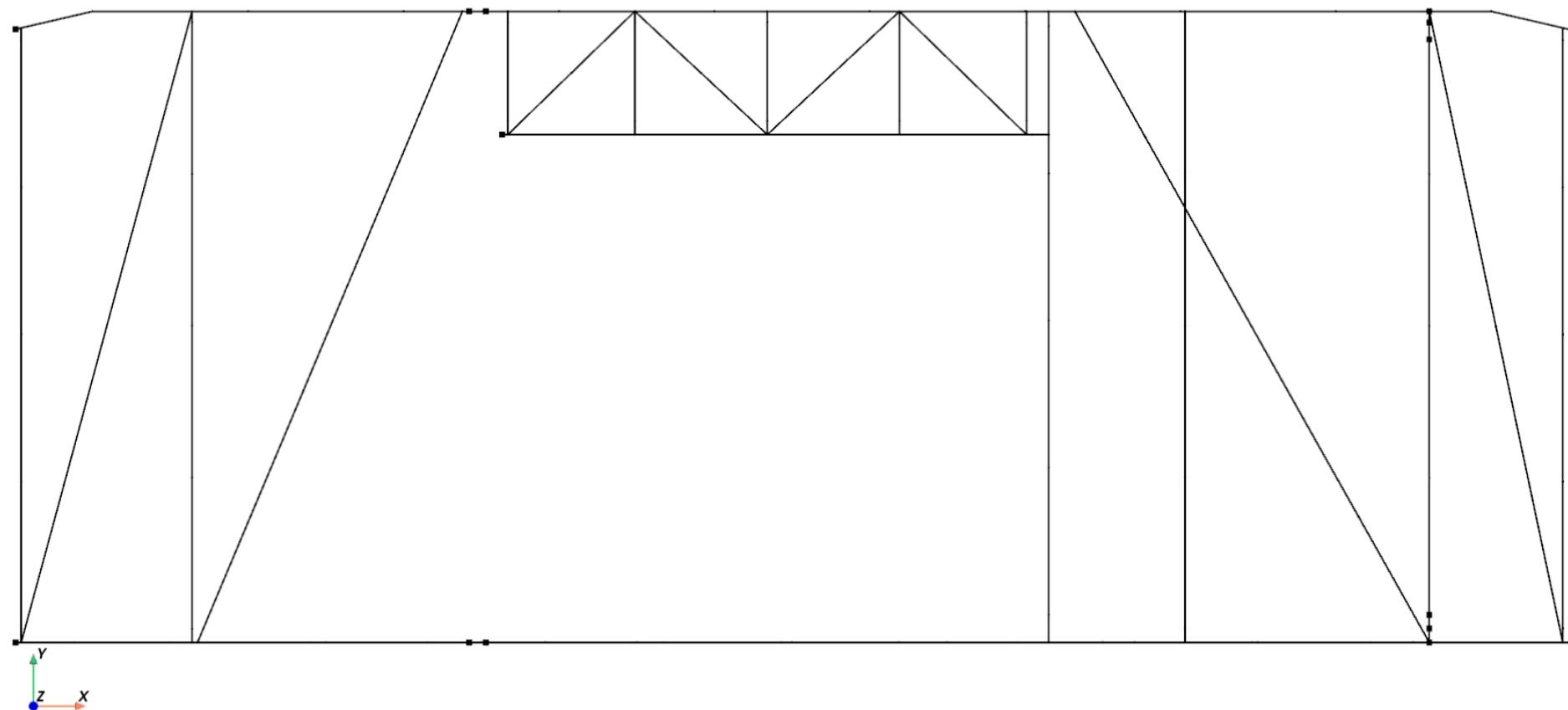
# TN complete PyNite solved network

RED: displacement x20 service\_patches



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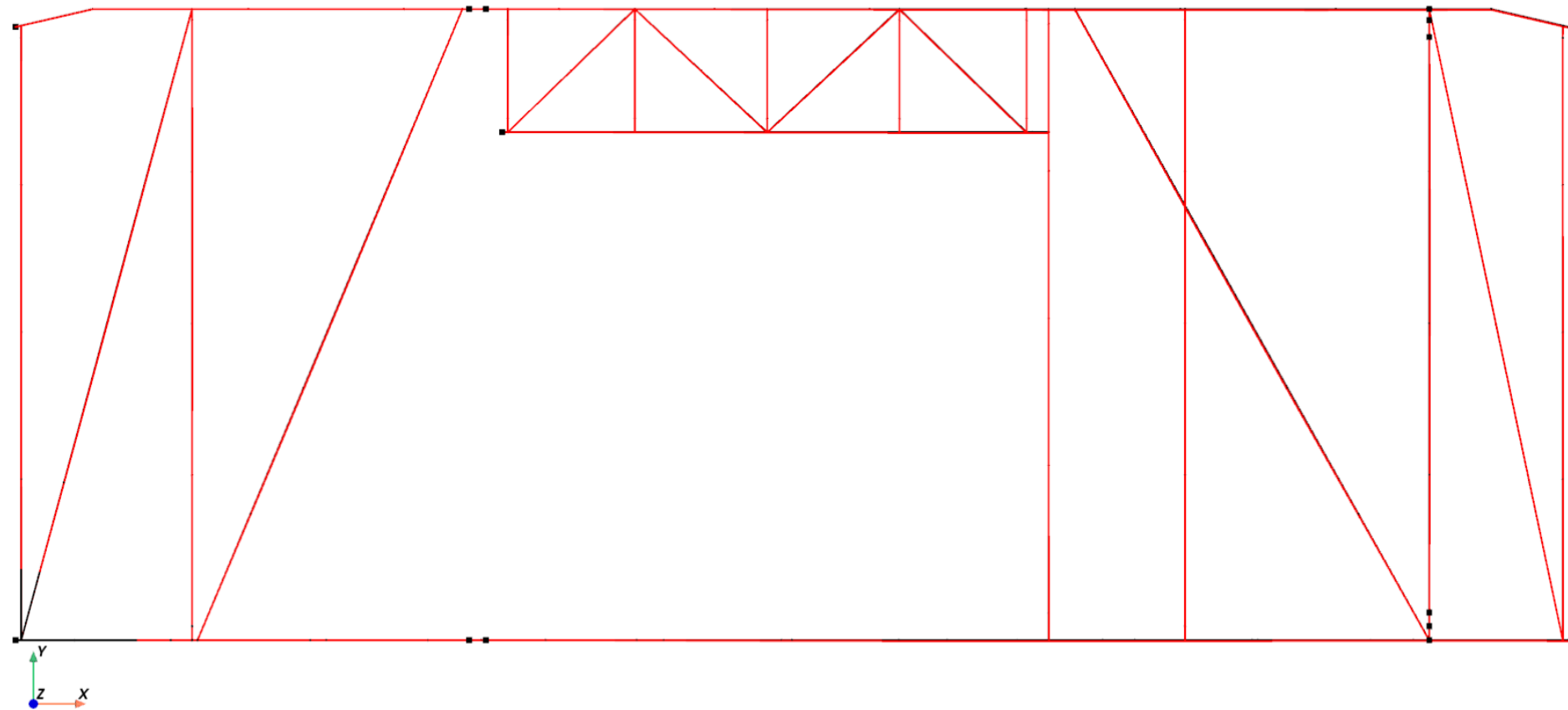
TN roof\_first PyNite solved network



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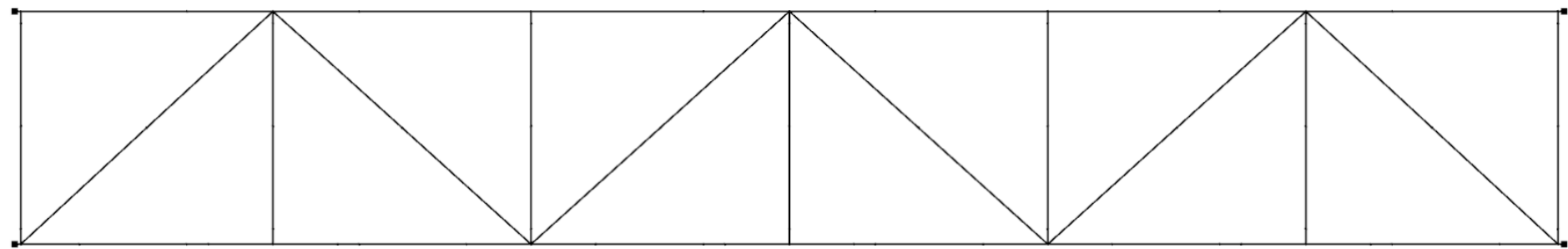
TN roof\_first PyNite solved network

RED: displacement x20 roof\_first\_service



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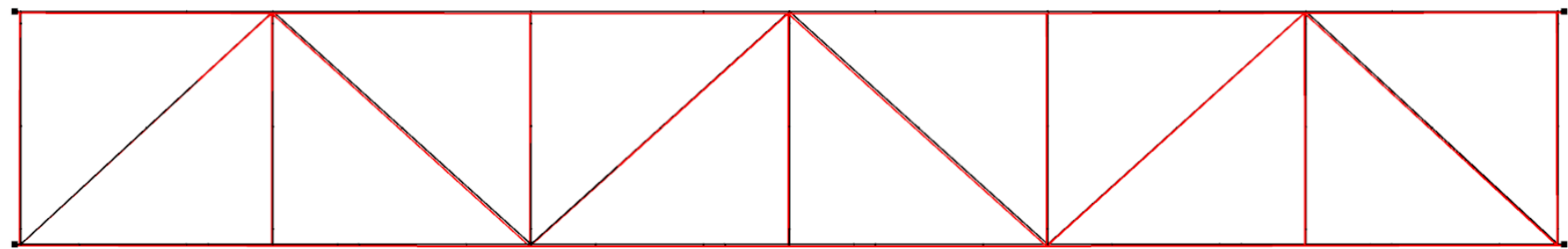
# TS complete PyNite solved network



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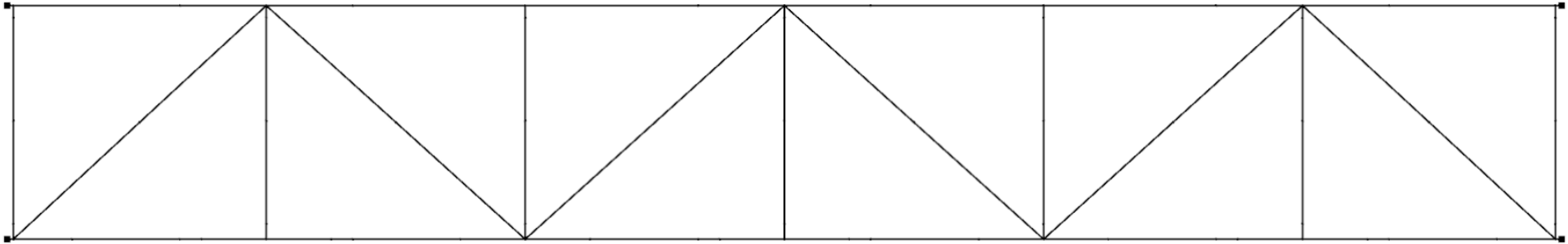
# TS complete PyNite solved network

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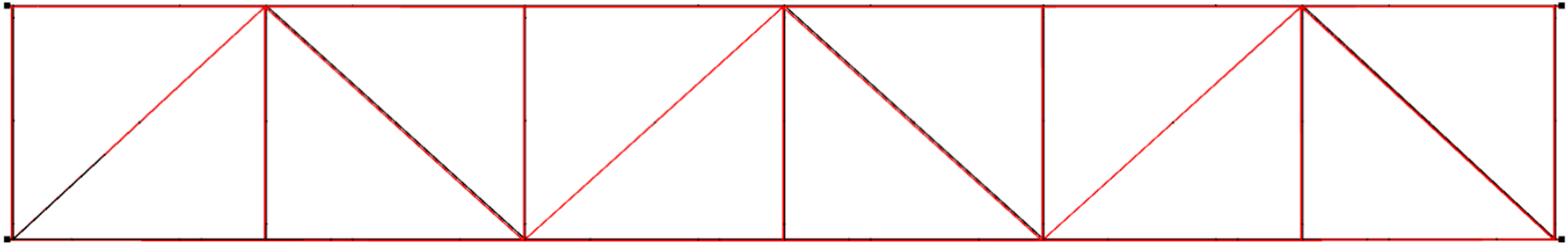
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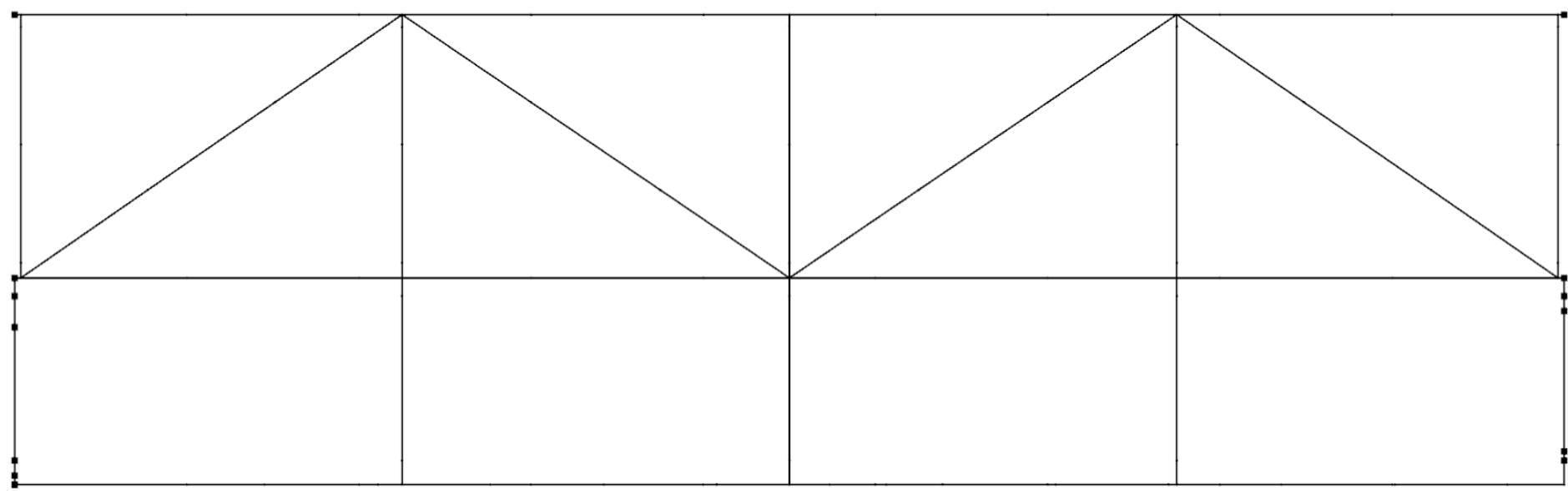
# TS roof\_first PyNite solved network

RED: displacement x20 roof\_first\_service



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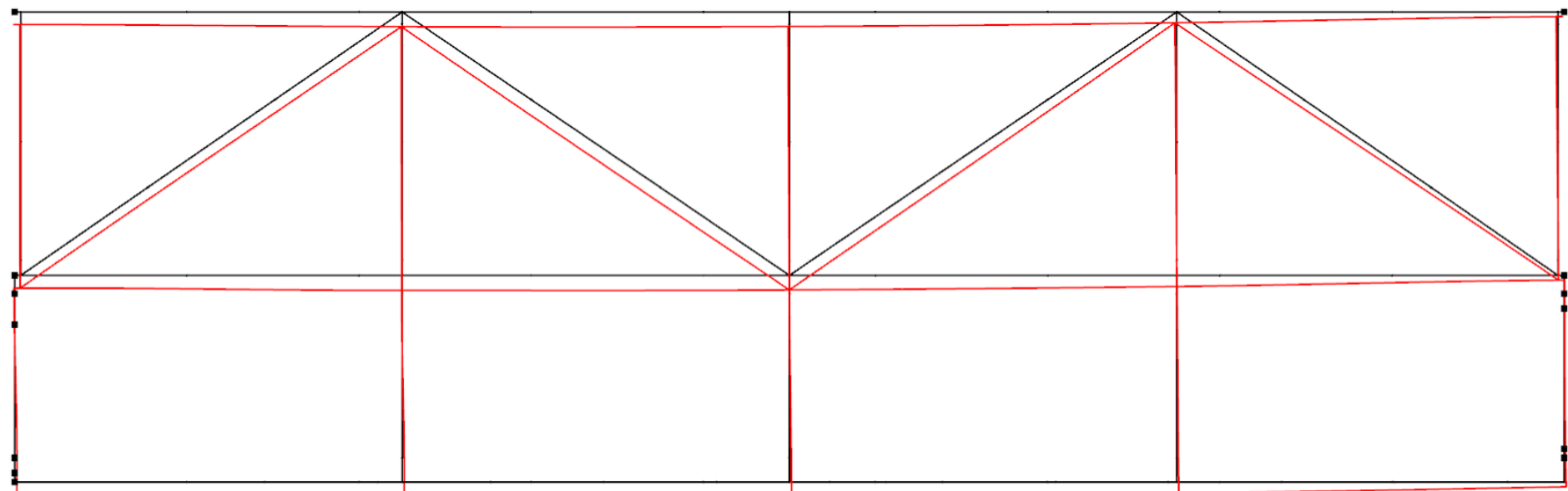
# T1 complete PyNite solved network



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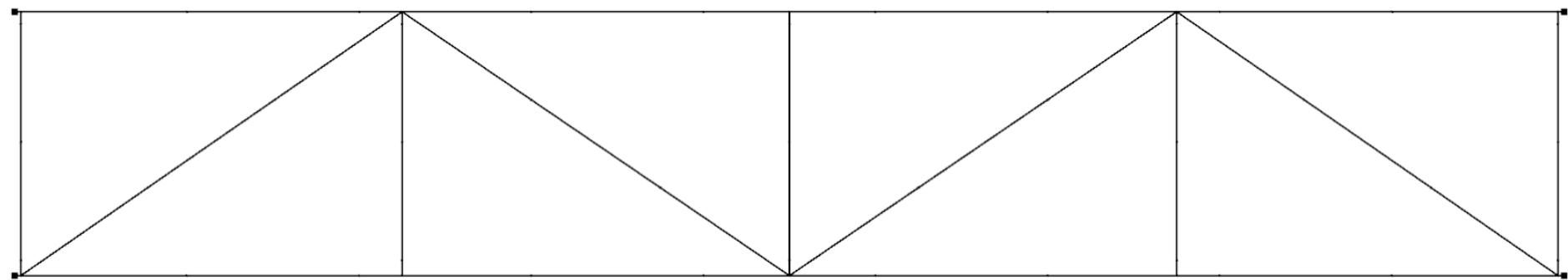
# T1 complete PyNite solved network

RED: displacement x20 service\_patches



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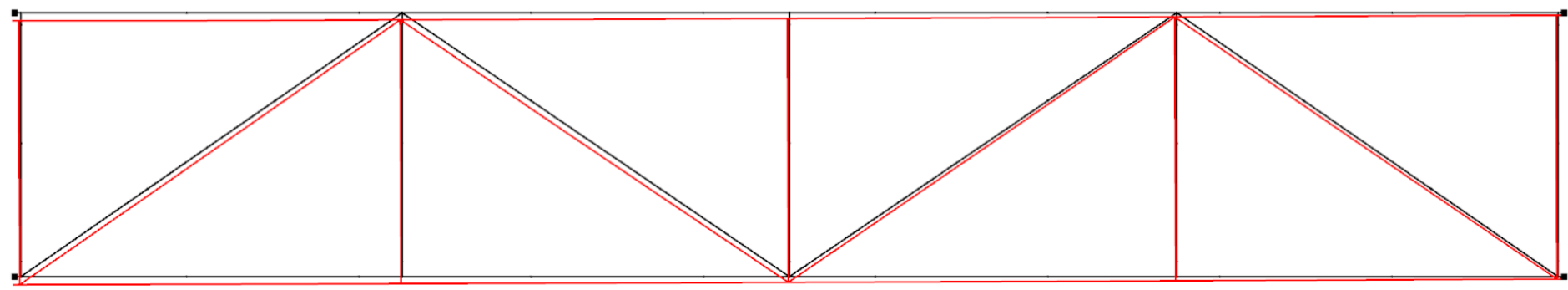
T1 roof\_first PyNite solved network



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T1 roof\_first PyNite solved network

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